THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** physical change	Physical.
2	**(b)** melting of ice	Melting.
3	**(d)** carbon dioxide	Fizz.
4	**(a)** chemical change	Chemical.
5	**(c)** burning of paper	Burning.
6	**(c)** carbon dioxide	CO_2.
7	**(c)** chemical change	Chemical.
8	**(b)** reddish-brown	Colour of rust.
9	**(a)** rusting	Slow.
10	**(a)** physical change	Physical.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No new substance (1); E.g. melting (1).	1+1
12	Iron oxide (1); Reddish-brown (1).	1+1
13	Can be undone (1); E.g. freezing (1).	1+1
14	New substance formed (1); E.g. burning (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Melting of ice (1); Boiling of water (1); Cutting paper (1).	1+1+1
16	Rain (1); Seasons (1); Day and night (1).	1+1+1
17	Wax remains wax (1); Solidifies again (1); No new substance (1).	1+1+1
	OR: Burning (1); Rusting (1); Cooking (1).	1+1+1
18	Cooking (1); Making a house (1); Making ice (1).	1+1+1
	OR: Gases (1); Ash (1); Heat and light (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Melting (1); Boiling (1); Freezing (1); Tearing paper (1); Dissolving sugar (1).	1+1+1+1+1
	OR: Add alum to hot water (1); Stir (1); Filter (1); Cool (1); Crystals form (1).	1+1+1+1+1
20	Burning (1); Rusting (1); Cooking (1); Digestion (1); Fermentation (1).	1+1+1+1+1
	OR: Rusting: slow (1); Ripening: slow (1); Cracker: fast (1); Match: fast (1); Balloon bursting: fast (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** water	Moisture.
2	**(b)** burning of a match	Fast.
3	**(d)** chemical change	Chemical.
4	**(d)** paint	Paint.
5	**(c)** rain	Natural.
6	**(b)** physical change	Physical.
7	**(b)** galvanisation	Galvanised iron.
8	**(c)** making a cake	Human action.
9	**(c)** physical change	Physical.
10	**(c)** periodic	Repeats at regular intervals.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Air (1); Water (1).	1+1
12	Cannot be undone (1); E.g. cooking (1).	1+1
13	Melting (1); Freezing (1).	1+1
14	Paint (1); Galvanise (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Melting (1); Freezing (1); Stretching a spring (1).	1+1+1
16	Painting (1); Oiling (1); Galvanising (1).	1+1+1
17	Iron nail (1); Iron gate (1); Steel tools (1).	1+1+1
	OR: Burning wood (1); Cooking (1); Ripening (1).	1+1+1
18	Colour change (1); Gas formed (1); Heat or light (1).	1+1+1
	OR: Salt (1); Sugar (1); Alum for water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	New substance (1); Reversibility (1); Energy change (1); Examples (1); Appearance (1).	1+1+1+1+1
	OR: Rain (1); Seasons (1); Cooking (1); Building (1); Painting (1).	1+1+1+1+1
20	Painting (1); Oiling (1); Galvanising (1); Stainless steel (1); Keep dry (1).	1+1+1+1+1
	OR: Wax melts (1); Wick burns (1); Wax vapour burns (1); Heat and light (1); Gases formed (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** saturated solution	Cooling.
2	**(c)** chemical change	Chemical.
3	**(b)** periodic change	Repeats daily.
4	**(a)** both physical and chemical changes	Wax melts and burns.
5	**(a)** chemical change	Chemical.
6	**(a)** periodic	Repeats monthly.
7	**(b)** chemical change	New substances.
8	**(b)** physical change	Physical.
9	**(c)** fast change	Quick.
10	**(c)** physical change	Recoverable.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Making crystals (1); From solution (1).	1+1
12	Rusting (1); Burning (1).	1+1
13	Coating with zinc (1); Protects iron (1).	1+1
14	Day and night (1); Moon phases (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Rusting (1); Ripening (1); Growth (1).	1+1+1
16	Only shape changes (1); No new substance (1); Pieces are the same material (1).	1+1+1
17	Prevent rust (1); Look good (1); Keep air and water away (1).	1+1+1
	OR: Burning a cracker (1); Lighting a match (1); Bursting a balloon (1).	1+1+1
18	New substances form (1); Heat and light (1); Cannot reverse (1).	1+1+1
	OR: Cooking (1); Melting butter (1); Curdling milk (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take three tubes (1); Put nails (1); Keep for a week (1); Look at nails (1); Only moist air tube rusts (1).	1+1+1+1+1
	OR: Take baking soda (1); Add vinegar (1); Fizzing (1); Gas formed (1); Chemical change (1).	1+1+1+1+1
20	Salt for food (1); Sugar (1); Alum (1); Gems (1); Blue vitriol (1).	1+1+1+1+1
	OR: Weakens iron (1); Damages bridges (1); Damages vehicles (1); Costs money (1); Prevent it (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.